SULIT

UTM UNIVERSITI TEKNOLOGI MALAYSIA Fakulti Komputeran

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER I 2022 / 2023

SUBJECT CODE	**: SECR2043 OPERATING SYSTEMS**
SECTION	**: SECR/J/B/V/P/ SCSR/J/B/V/P SEEL/H**
TIME	**: 2.30 PM to 5.30 PM**
DATE/DAY	**: 20 FEBRUARY 2023**
VENUES	**:**

INSTRUCTIONS:

Answer all questions from PART A and PAPRT B. Show all your works.

This test will contribute **35%** towards the total marks of 100%.

(Please Write Your Lecturer Name and Section)

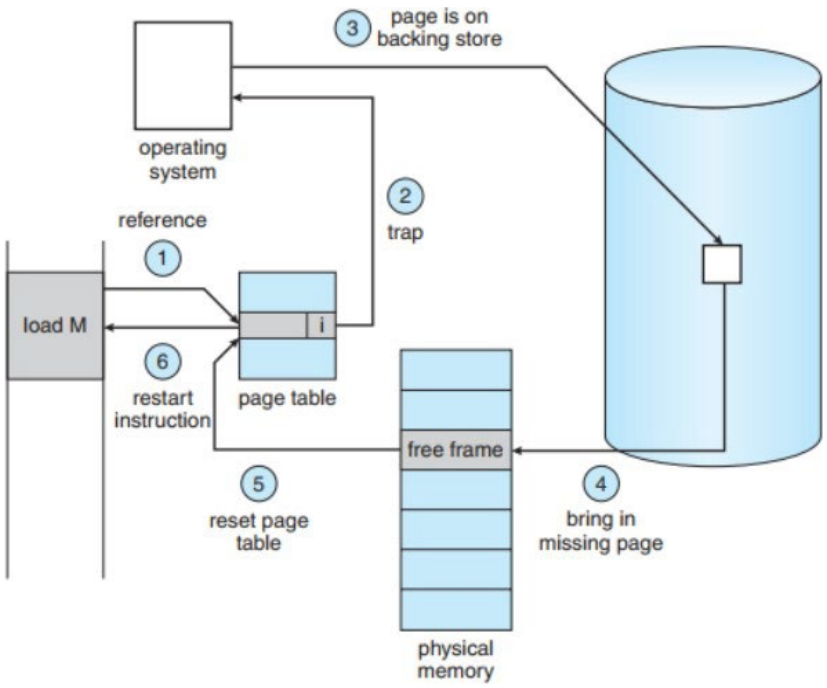
Name	
Matric No.	
Year / Course	
Section	**[01] [02] [03] [04] [05] [15]**
Lecturer Name	**[DR FOAD] [DR AIDA] [DR ZURIAHATI] [DR MADNI] [DR SHAHIDA]**

This question booklet consists of 12 pages including the front page.

PART A: OBJECTIVES

Answer ALL questions.

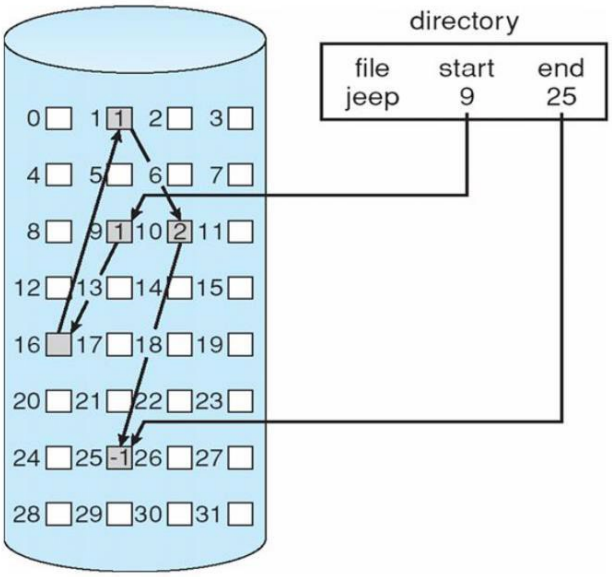
1.	In which situation does a page fault occurs?
A.	The page is not present in main memory
B.	The page is corrupted by application software
C.	The page is in present in main memory
D.	None of the above
2.	A computer's performance usually becomes better by increasing the RAM because _____.
A.	fewer segmentation fault occurs
B.	virtual memory increases
C.	larger RAMs are fasterf
D.	fewer page faults occur
3.	Dirty bit is used to show the _____.
A.	page that is modified after being loaded into cache memory
B.	wrong page
C.	page with low frequency occurrence
D.	page with corrupted data
4.	1. A _____ is a series of code sections can be loaded in memory and execute.
A.	A. Text file
B.	B. Source file
C.	C. Executable file
D.	D. Sequence file
5.	File operations activities will include the following, **EXCEPT**:
A.	Write
B.	Delete

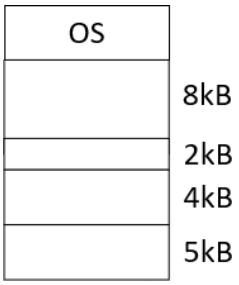
C.	Truncate
D.	Hanging
6.	Based on Figure 1, determine the **TRUE** sequence and the actions the operating system takes when a page fault occurs.  **Figure 1** Steps in handling a page fault
	I. If the reference was invalid, the process is terminated. Otherwise, the page must be paged in. II. The instruction that caused the page fault must now be restarted from the beginning. III. A disk operation is scheduled to bring in the necessary page from disk. IV. The memory address requested is first checked, to make sure it was a valid memory request. V. When the I/O operation is complete, the process's page table is updated with the new frame number to indicate that the page is now in memory. VI. A free frame is located, possibly from a free-frame list.
A.	I, VI, III, IV, V, II
B.	III, I, III, VI, II, IV
C.	IV, I, VI, III, V, II

D.	I, II, III, IV, V, VI
7.	The physical address space is conceptually divided into a number of fixed-size blocks, known as _____.
A. B. C. D.	Frames segments Pages None of the above
8.	Which of the following are attributes that are commonly used in file attributes? I. Name II. Identifier III. Type IV. Size
A. B. C. D.	A. I and II B. II and III C. I, III and IV D. All of the above
9.	A _____method allow programs to read and write records rapidly in a numbered sequence of blocks or records.
A. B. C. D.	A. sequential access B. direct access C. index sequential access D. next access
10.	A process loses its ability to use the CPU when it's swapped out of the memory. Determine another situation where a process loss its ability to use the CPU without being swapped out from the CPU.
A. B. C. D.	a) **When an interrupt occurs** b) When there is a free space in main memory c) When CPU is in the idle mode d) When generating logical address

	e)
	f)
11.	The following concept file describes a _____? Listings of filenames and their attributes Attributes are information related to a files Organize files into a logical structure
A. B. C. D.	A. database B. directory C. field D. record
12.	File systems provide _____to file storage and the mapping of _____to physical address.
A. B. C. D.	A. user interface, memory address B. random access, logical address C. user interface, logical address D. random access, physical block
13.	Which of the followings are the THREE allocation methods for disk space? I. Linked II. Indexed III. Contiguous IV. Sequential
A. B. C. D.	A. I, II and III B. II, III and IV C. I, III and IV D. I, II and IV
14.	What is the available solution to reduce fragmentation?
A. B. C. D.	a) Free the allocated memory space b) Remove the process from memory space c) Change the physical address space to be non-contiguous d) Change the logical address space to be contiguous

	e)
	f)
15.	Which of the followings are **TRUE** about Modified Contiguous Allocation scheme? I. This scheme is used to solve the problem of internal fragmentation II. For this scheme, initially, a contiguous block of space is allocated III. If the initial block allocated is not enough, another contiguous block, known as extent, is added IV. When using this scheme, a file may consist of one or more extent blocks
A. B. C. D.	A. I, II and III B. II, III and IV C. I, III and IV D. All of the above
16.	Which of the followings are **TRUE** about indexed allocation method? I. To find the and read the *ith* block, the pointer in the *ith* index-block entry is used II. Indexed allocation method allows random access III. The index block size dilemma can be solved using Index Pointer Structure IV. An index table is needed for indexed allocation method
A. B. C. D.	A. I, II and III B. II, III and IV C. I, II and IV D. All of the above
17.	What are the cases that lead to dynamic partition? I. When the holes are adjacent II. When the free blocks are separated by a non-free block III. When the free blocks are scattered IV. When there is free space in memory
A. B. C. D.	a) I, II and III b) II and III c) II, III and IV d) I, II, III and IV

18.	1. Which of the followings are **TRUE** about Figure 2?  **Figure 2:** An **X** scheme for disk allocation method I. The X scheme in Figure 2 above refer to linked allocation method II. Each block contains a pointer to the next block III. Each file stored using the X scheme is a linked list of blocks IV. Each file stored using the X scheme has its own index blocks of pointers to its data blocks
A. B. C. D.	A. I, II and III B. II, III and IV C. I and III D. I, II and IV
19.	A deadlock avoidance algorithm directly examines the _____ to ensure the circular wait condition will never occur.
A. B. C. D.	Resource Allocation State Resource Available State Resource Need State Resource Utilization State

20.	Deadlock is more precisely described in term of directed graph known as:
A. B. C. D.	Resource Allocation Graph Resource Scheduling Graph Deadlock Detection Graph Deadlock Allocation Graph
21	Choose the **TRUE** statements related to non-contiguous memory allocation. I. Each process contained in a single contiguous section of memory II. Each process contained in non-sequential location of memory III. Segmentation IV. Dynamic partitions
A. B. C. D.	a) I and III b) **II and III** c) III and IV d) I and IV
22	In deadlock prevention, the mutual exclusion condition occurs if:
A. B. C. D.	at least one resource must be non-shareable all resources must be shareable when a process requests a resource, it does not hold any other resource all of the above
23.	Based on Figure 3, determine the total internal fragmentation after contiguous fixed memory allocation of `Job 1=7kB, Job 2=2kB, Job 3=4Kb, Job 4=2kB`.  **Figure 3**
A. B.	a) **4kB** b) 3kB

C.	c) 1kB
D.	d) 5kB
24.	A request queue 17, 102, 45, 80. Head pointer at 70. Which request that will first satisfy, if using **SCAN** inner to outer?
A.	45
B.	102
C.	80
D.	0
25.	A system is in a safe state only if there exists a:
A.	Safe Allocation
B.	Safe Resource
C.	Safe Sequence
D.	Safe Process
26.	The following are the disk scheduling algorithms, **EXCEPT;**
A.	SSTF
B.	C-SCAN
C.	LRU
D.	LOOK
27.	A request queue 17, 102, 45, 80. Head pointer at 70. Which request that will third satisfy, if using **SSTF**?
A.	80
B.	102
C.	45
D.	17
28.	Which of the following statement is **FALSE**.
A.	Disk striping can increase the data transfer speed due to parallel transfer.
B.	Mirroring will duplicate each disk.

C.	Mirroring is similar to shadowing.
D.	Data striping make data retrieval from disk slow.
29.	Disk scheduling (in cylinders) not allow data to be accessed from _____.
A.	Outer to inner
B.	Top to bottom
C.	Inner to outer
D.	Inner to outermost
30.	A set of Processes = {P1, P2, P3, ..., Pn} is waiting for resources in such a way that P1 → R1 → P2 → R3 → P3 → R2.. Pn → Rk → P1. This condition of deadlock is referred as: I. Mutual Exclusion II. Hold and Wait III. Circular Wait IV. Non-Preemption
A.	I and II
B.	II and III
C.	I and III
D.	III and IV

PART B: STRUCTURED

Answer ALL questions in answer booklet.

QUESTION 1 [14 MARKS]

1. One of the techniques for a file system to maintain free-space list in order to track available blocks is through bit map. Assume that there are 4 bits per word and the bit map is as in the Table 1. Show how the address of the first free block is calculated. [2]

Block No	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	…
Bit map	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	…
Offset	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	…

Table 1: Bit map for free space management

2. Figure 1 illustrates the disk space containing `file`, `count`, `tr`, `mail`, `list` and `f` stored in some blocks contiguously.

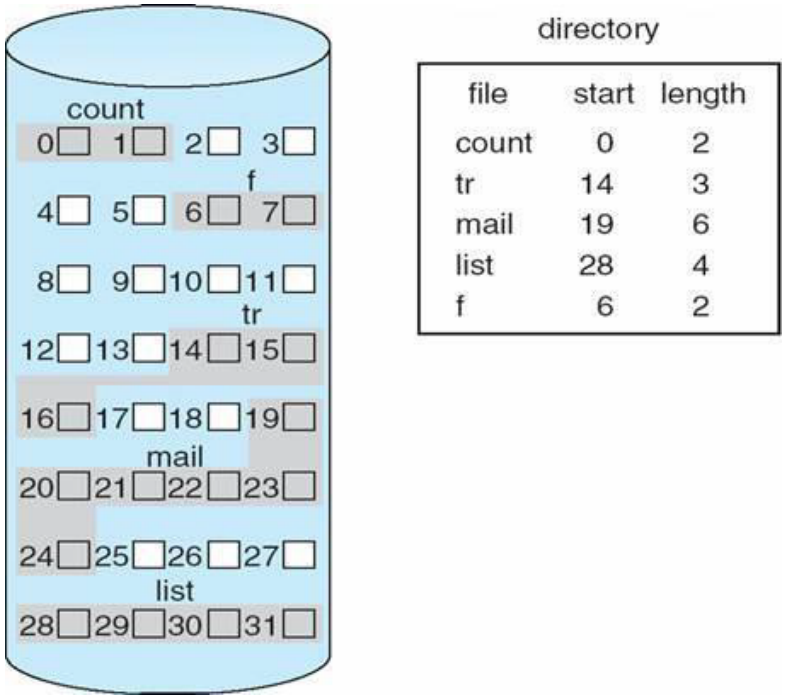


Figure 1: Files stored contiguously on a disk

a) Assuming that each block can only store 4 block address, show how the free space is represented using Grouping technique. Draw your answer and label everything [5]

b) From your answer in 2a), determine which blocks become index blocks [1]

c) From your answer in 2a), determine which blocks become data blocks [2]

d) From Figure 1 above, show how the free space is represented using Counting technique. List your answer [4]

QUESTION 2 [11 MARKS]

A request queue is 89, 129, 73, 22, 140, 46, 0, 7. Head point at 100. Outer to inner

a) Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests using SCAN. Draw the head movement. [4]

b) Calculate the total distance (in cylinders) that the disk arm moves to satisfy the requests using C-SCAN. Draw the head movement. [4]

c) Please suggest ONE disk scheduling that can outperformed the performance of SCAN and c-SCAN. Why? [3]

QUESTION 3 [9 MARKS]

Given the following reference string:

5	**2**	**0**	**3**	**6**	**3**	**0**	**1**	**2**	**0**	**6**	**0**	**3**	**1**	**2**	**3**	**4**	**1**	**2**

Calculate how many page faults will occur:

a) If **OPT page replacement** with 4-page frames is used. Show all your workings. [4]

b) If **LRU page replacement** with 4-page frames is used. Show all your workings. [4]

c) From the above results, which has the best performance and discuss? [1]

QUESTION 4 [15 MARKS]

1. Figure 2 shows the physical memory of 5 segments (Segment A – E). Gray boxes refer to used blocks. Limit for each segment is listed in Table 2. Solve the following questions.

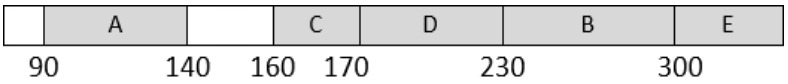


Figure 2: Physical memory

Table 2: Segment table

Segment	Limit
A	50
B	70
C	10
D	60
E	15

a) State the base address of **Segment C** and **E**. [1]

b) Determine the physical memory address of B:6 and E:16. Show all your workings and justify your answers. [4]

2. Consider the physical address size of 64 bytes and page size is 4 bytes.

a) Determine the page number of a physical address and page offset of a logical address [2]

b) How many frames for the physical memory? [1]

c) Assume that the logical addresses in logical memory are mapped in increment numerical order starting with 0. List the logical addresses for pages 2 and 4. [3]

d) Next, based on the answer from Q(c) and Table 3, get the offset for the logical memory location of **CC**. [2]

e) Based on the answer from Q(d) and Table 3, calculate the physical address in binary for **CC**. [2]

Table 3

Page	Frame	Content
⋮		⋮
⋮		⋮
2	8	AA
		BB

		CC
		DD

QUESTION 5 [21 MARKS]

Given the following information

- 5 processes (P_0, P_1, P_2, P_3, P_4) arrived at t_0.
- Resources (A-9 instances, B-5 instances and C-9 instances)
- Available (2, 2, 3)
- Show all the calculation steps in detail

a) What is the safe state sequence produced? [9]

b) If P_2 requests for (1, 0, 2), then find the safe state sequence. [10]

c) If P_3 requests for (4, 3, 3), then find the safe state sequence. [2]

Processes	**Allocation**			**Maximum**		
	A	**B**	**C**	**A**	**B**	**C**
P_0	0	2	0	3	5	7
P_1	2	0	0	2	2	3
P_2	3	0	3	2	0	9
P_3	0	0	2	3	3	4
P_4	2	1	1	2	2	2